

Human-Supervised Autonomous Multi-Agent Systems

VR and Desktop Command Interface Evaluation

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Autonomous Agents Are Powerful – But Humans Still Matter

Current agent architectures force a binary choice: full autonomy with no strategic flexibility, or direct control that fails to scale. Neither extreme works in dynamic, complex environments.

Full Autonomy ✗

- No human strategic input
- Limited adaptability
- Rigid pre-programmed rules

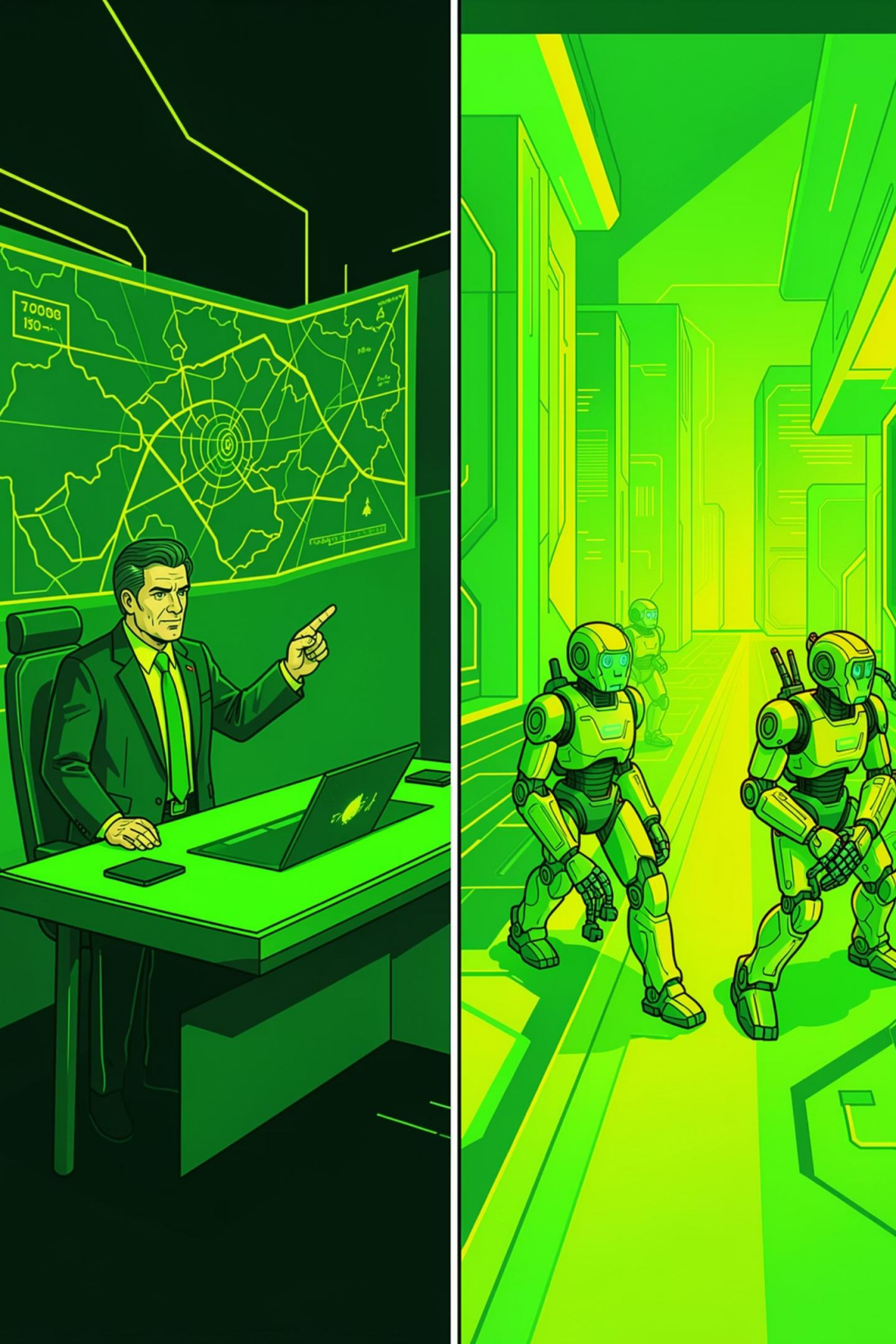
Direct Control ✗

- Doesn't scale for multiple agents
- Wastes agent intelligence
- Micromanagement required

Strategic Supervision ✓

- Balanced human-agent collaboration
- Human guides, agents execute
- Scales effectively across teams

Key Question: How should humans supervise autonomous agent teams – without undermining their intelligence?



Strategic-Tactical Separation

The core architectural principle: keep human cognition at the strategic layer, and preserve agent autonomy at the tactical layer.

Human – Strategy

- Where agents should move
- Which areas to cover or hold
- When to regroup or release

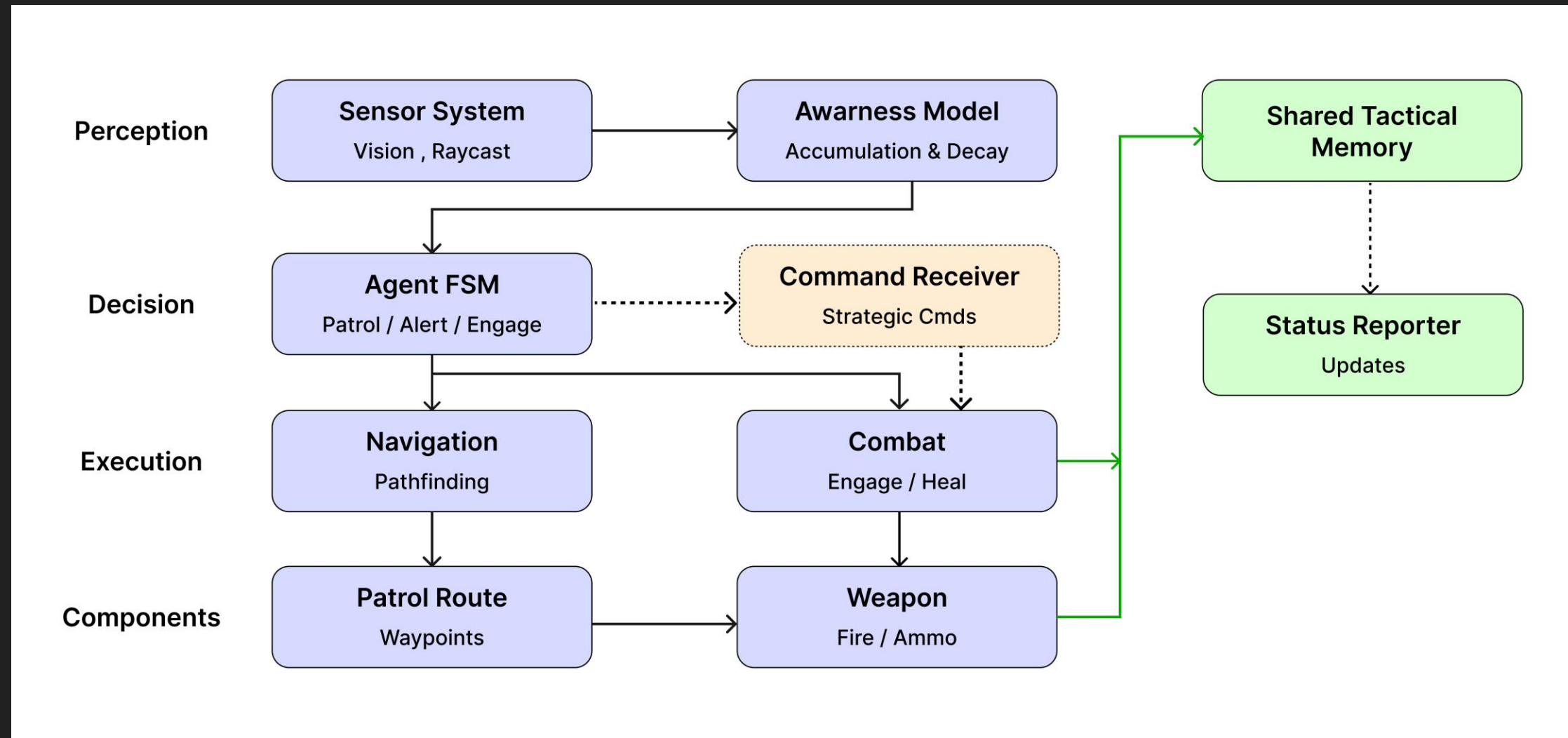
Agents – Tactics

- When to detect and engage enemies
- When to shoot, heal, or reload
- How to coordinate with teammates

Human gives strategy. Agents handle tactics.

Command Integration Architecture

A layered architecture ensures human commands are interpreted at the right level of abstraction — never overriding agent tactical decision-making.



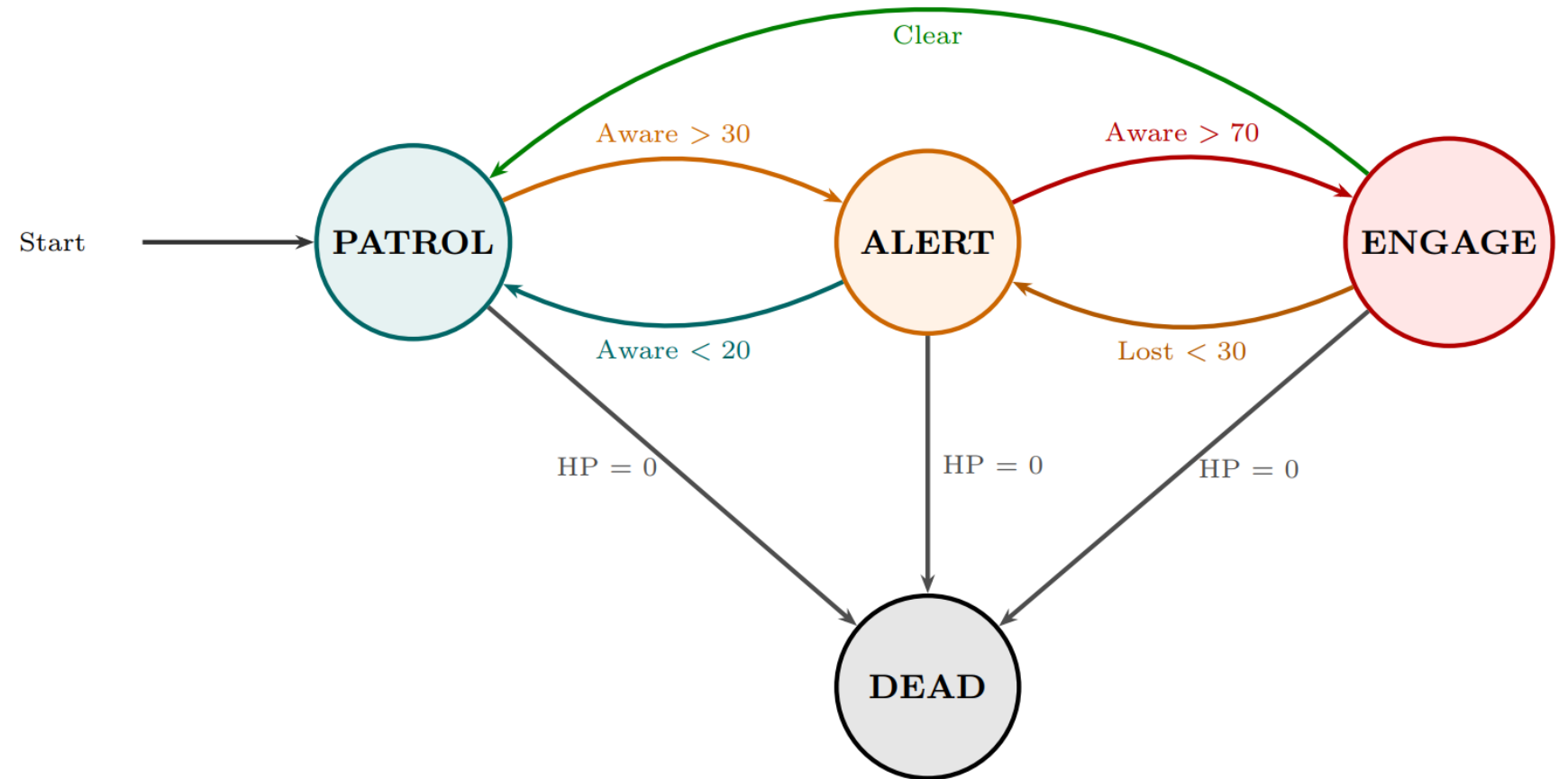
Strategic commands only. Tactical execution remains fully autonomous.

Each Agent Thinks with a Finite State Machine

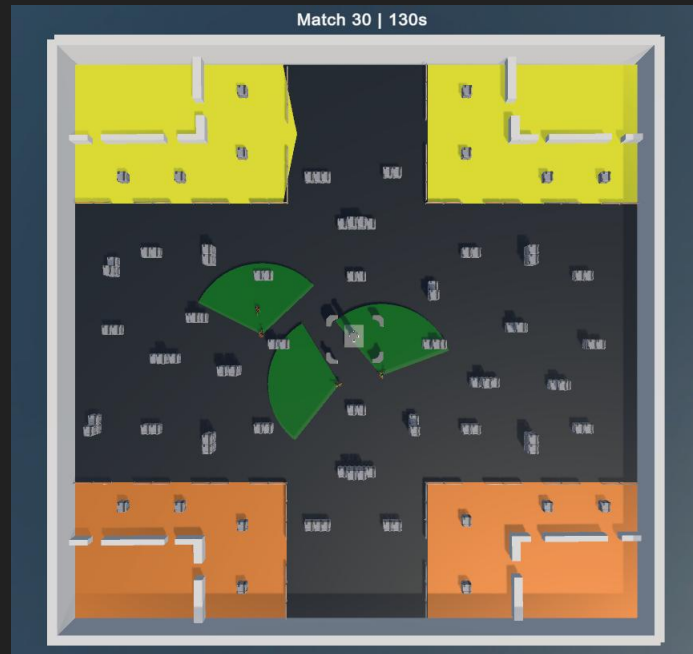
Agents operate independently through four core states, transitioning based on real-time awareness and health thresholds.

Autonomous Capabilities

- Vision-based enemy detection
- Combat execution — shoot, heal, reload
- Team threat broadcasting
- Pathfinding and navigation



Same Agents. Same Commands. Different Viewpoint.



Desktop Interface

- Overhead operations view
- Superior spatial overview
- Positioning-focused commands
- Strategic placement emphasis



VR Interface

- Ground-level immersive perspective
- Human-scale situational awareness
- Reactive coordination style
- Active intervention commands

Both interfaces use identical commands: **Move To** · **Regroup** · **Hold** · **Release**



300 Systematic Trials

Autonomous Baseline

100 trials — both factions fully autonomous, zero human intervention

Desktop-Supervised

100 trials — Alpha team commanded via desktop, Beta adversary autonomous

VR-Supervised

100 trials — Alpha team commanded via VR headset, Beta adversary autonomous

Configuration: 3 vs 3 agents · 5-minute limit · Urban terrain · Randomized spawns

Metrics: Win rate, casualties, trial duration, command acceptance, agent behavior fidelity

VR Achieved the Highest Favorable Outcome Rate

39%

Autonomous
Baseline win rate

41%

Desktop
+2% vs baseline

56%

VR
+17% vs baseline ✨

Trial Duration



Autonomous
142.2 seconds



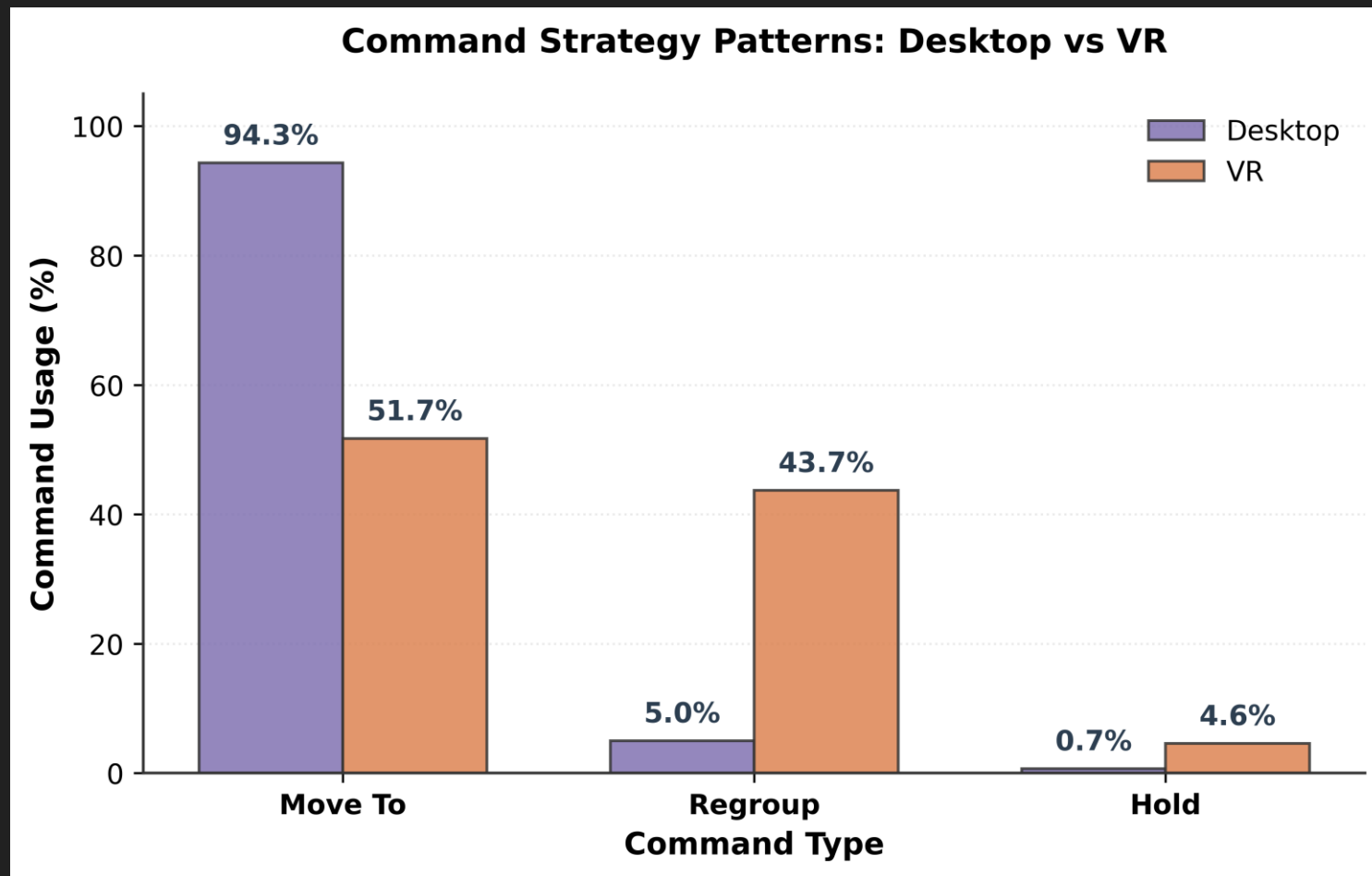
Desktop
124.5 sec — 13% faster



VR
102.9 sec — 28% faster

✔ **Key Finding:** VR supervision delivers substantial performance improvement — +17% win rate over the autonomous baseline.

Desktop and VR Produced Distinct Supervision Styles



Desktop – Strategic Positioning

94.7% Move To commands · Commands issued during safe periods · Acceptance rate: **92.4%**

VR – Active Coordination

57.5% Move To · 32.8% Regroup · Commands during live combat · Acceptance rate: **73.7%**

📄 Interface perspective shapes supervision behavior — not just command preferences.

The Autonomy–Performance Trade-off

Desktop and VR supervision optimize for fundamentally different objectives — neither is universally superior.

Desktop – Autonomy Preservation ✓

- Agent behavior closely mirrors baseline
- State transitions: 4.0 vs 3.9 baseline
- Engagement time: 6.3% vs 6.9% baseline
- Modest performance gain: **+2%**

VR – Performance Optimization ✓

- Highest win rate across all conditions
- State transitions: 5.5 vs 3.9 baseline
- Engagement time: 9.4% vs 6.9% baseline
- Substantial performance gain: **+17%**

Neither is "better" — they optimize for different objectives. Choose Desktop for behavioral fidelity. Choose VR for mission performance.

Demo



We extended the research system into a mission-style simulation with:

VR player

Shooting and health

Objective interaction

Mission timer

Win/loss conditions

THANK YOU